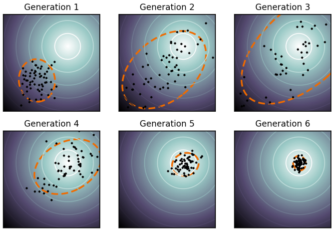
Optimization in Machine Learning

Evolutionary Algorithms
CMA-ES Algorithm



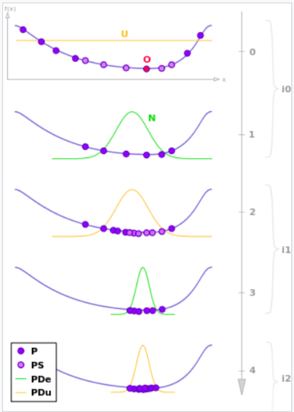
Learning goals

- CMA-ES strategy
- Estimation of distribution
- Step size control

ESTIMATION OF DISTRIBUTION ALGORITHM

Instead of population, maintain distribution to sample offspring from

1. Draw λ offspring $\boldsymbol{x}^{(i)}$ from $p(\cdot|\boldsymbol{\theta}^{[t]})$
2. Evaluate fitness $f(\boldsymbol{x}^{(i)})$
3. Update $\boldsymbol{\theta}^{[t+1]}$ with μ best offspring



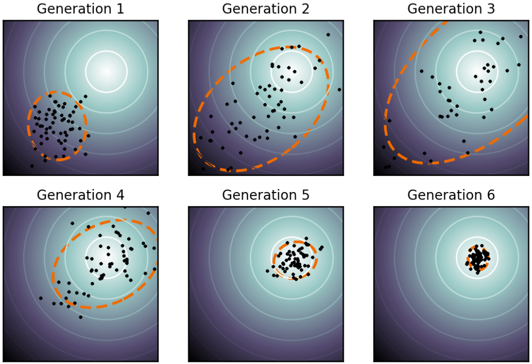
Estimation of distribution algorithm. For each iteration *i*, a random draw is performed for a population *P* in a distribution *PDu*. The distribution parameters *PDe* are then estimated using the selected points *PS*. The illustrated example optimizes a continuous objective function *f(X)* with a unique optimum *O*. The sampling (following a normal distribution *N*) concentrates around the optimum as one goes along unwinding algorithm.

COVARIANCE MATRIX ADAPTATION

- Sample distribution is multivariate Gaussian

$$\boldsymbol{x}^{[t+1](i)} \sim \mathbf{m}^{[t]} + \sigma^{[t]}\mathcal{N}(\mathbf{0}, \mathbf{C}^{[t]}) \quad \text{for } i = 1, \dots, \lambda$$

- $\boldsymbol{x}^{[t+1](i)} \in \mathbb{R}^d$ i-th offspring; $\lambda \geq 2$ number of offspring
- $\mathbf{m}^{[t]} \in \mathbb{R}^d$ mean value and $\mathbf{C}^{[t]} \in \mathbb{R}^{d \times d}$ covariance matrix
- $\sigma^{[t]} \in \mathbb{R}_+$ "overall" standard deviation/step size



- **Question:** How to adapt $\mathbf{m}^{[t+1]}$, $\mathbf{C}^{[t+1]}$, $\sigma^{[t+1]}$ for next generation $t+1$?

CMA-ES: BASIC METHOD - ITERATION 1

0. Initialize $\mathbf{m}^{[0]}, \sigma^{[0]}$ problem-dependent and $\mathbf{C}^{[0]} = \mathbf{I}_d$

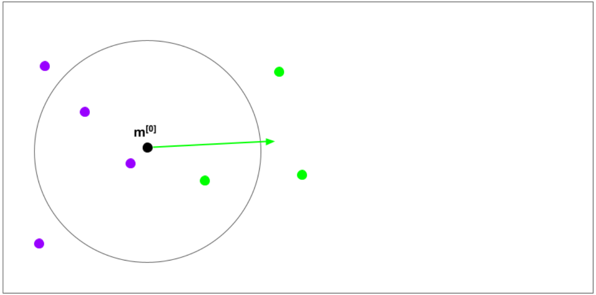
CMA-ES: BASIC METHOD - ITERATION 1

1. **Sample** λ offspring from distribution

$$\boldsymbol{x}^{[1](i)} = \mathbf{m}^{[0]} + \sigma^{[0]}\mathcal{N}(\mathbf{0}, \mathbf{C}^{[0]})$$

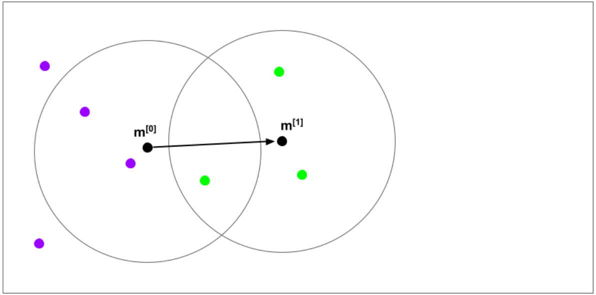
CMA-ES: BASIC METHOD - ITERATION 1

2. **Selection and recombination** of $\mu < \lambda$ best-performing offspring using fixed weights $w_1 \geq \ldots \geq w_\mu > 0, \sum_{i=1}^{\mu} w_i = 1$. $\boldsymbol{x}_{i:\lambda}$ is i-th ranked solution, ranked by $f(\boldsymbol{x}_{i:\lambda})$.
   Calculation of auxiliary variables ($\mu = 3$ points)
   $\boldsymbol{y}_w^{[1]} := \sum_{i=1}^{\mu} w_i(\boldsymbol{x}_{i:\lambda}^{[1]} - \mathbf{m}^{[0]})/\sigma^{[0]} := \sum_{i=1}^{\mu} w_i \boldsymbol{y}_{i:\lambda}^{[1]}$



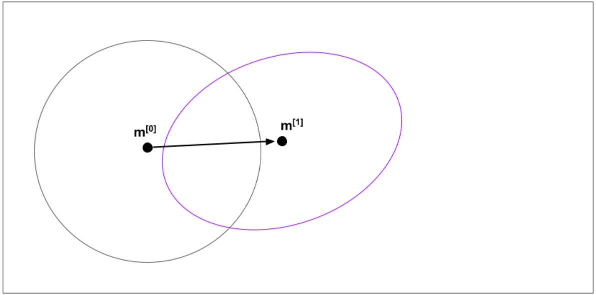
CMA-ES: BASIC METHOD - ITERATION 1

3. **Update mean**
   Movement towards the new distribution with mean $\mathbf{m}^{[1]} = \mathbf{m}^{[0]} + \sigma^{[0]} \boldsymbol{y}_w^{[1]}$.



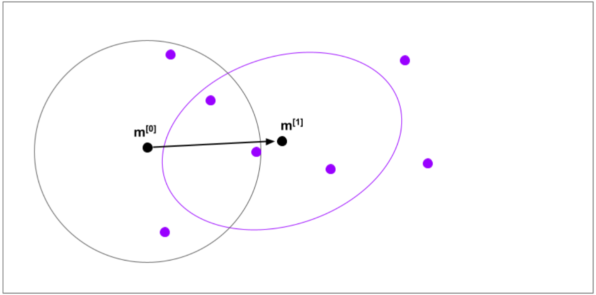
CMA-ES: BASIC METHOD - ITERATION 1

3. **Update covariance matrix** Roughly: elongate density ellipsoid in direction of successful steps.
   - $\mathbf{C}^{[1]}$ reproduces successful points with higher probability than $\mathbf{C}^{[0]}$.
   - Update $\mathbf{C}^{[0]}$ using sum of outer products and parameter c_μ:
     $\mathbf{C}^{[1]} = (1 - c_\mu)\mathbf{C}^{[0]} + c_\mu \sum_{i=1}^{\mu} w_i \boldsymbol{y}_{i:\lambda}^{[1]} (\boldsymbol{y}_{i:\lambda}^{[1]})^T$ (rank-μ update).



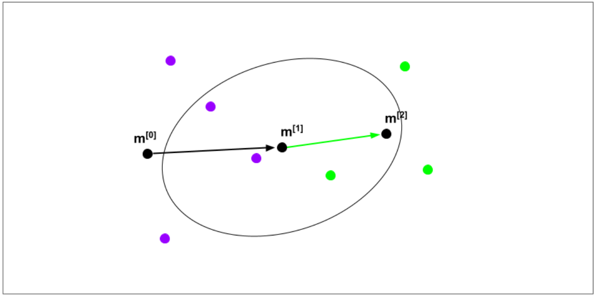
CMA-ES: BASIC METHOD - ITERATION 2

1. **Sample** from distribution for new generation



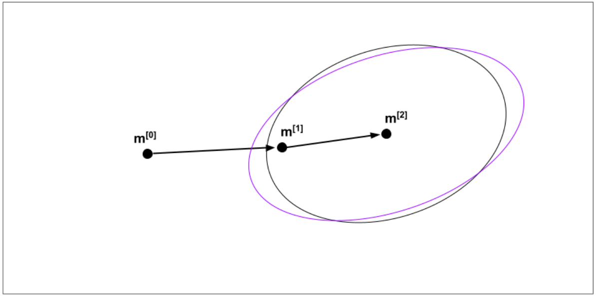
CMA-ES: BASIC METHOD - ITERATION 2

2. **Selection and recombination** of $\mu < \lambda$ best-performing offspring
3. **Update mean**



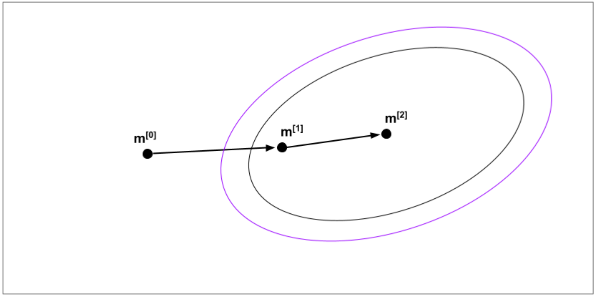
CMA-ES: BASIC METHOD - ITERATION 2

3. **Update covariance matrix**



CMA-ES: BASIC METHOD - ITERATION 2

3. **Update step-size** exploiting correlation in history of steps:
   - steps point in similar direction $\Longrightarrow$ increase step-size
   - steps cancel out $\Longrightarrow$ decrease step-size



UPDATING C: FULL UPDATE

Full CMA update of **C** combines rank-μ update with a rank-1 update using exponentially smoothed evolution path $\mathbf{p}_c \in \mathbb{R}^d$ of successive steps and learning rate c_1:

$$\mathbf{p}_c^{[0]} = \mathbf{0}, \quad \mathbf{p}_c^{[t+1]} = (1 - c_1)\mathbf{p}_c^{[t]} + \sqrt{\frac{c_1(2 - c_1)}{\sum_{i=1}^{\mu} w_i^2}} \boldsymbol{y}_w$$

Final update of **C** is

$$\mathbf{C}^{[t+1]} = (1 - c_1 - c_\mu \textstyle\sum_j w_j)\mathbf{C}^{[t]} + c_1 \underbrace{\mathbf{p}_c^{[t+1]}(\mathbf{p}_c^{[t+1]})^T}_{\text{rank-1}} + \underbrace{c_\mu \sum_{i=1}^{\mu} w_i \boldsymbol{y}_{i:\lambda}^{[t+1]}(\boldsymbol{y}_{i:\lambda}^{[t+1]})^T}_{\text{rank-}\mu}$$

- Correlation between generations used in rank-1 update
- Information from entire population is used in rank-μ update

UPDATING σ: METHODS STEP-SIZE CONTROL

- $1/5$**-th success rule**: increases the step-size if more than 20 % of the new solutions are successful, decrease otherwise

- σ**-self-adaptation**: mutation is applied to the step-size and the better - according to the objective function value - is selected

- **Path length control via cumulative step-size adaptation (CSA)**
  Intuition:
  - Short cumulative step-size $\triangleq$ steps cancel $\rightarrow$ decrease $\sigma^{[t+1]}$
  - Long cumulative step-size $\triangleq$ corr. steps $\rightarrow$ increase $\sigma^{[t+1]}$